A Precision-Gated Evidence–Judge Cascade for Detecting Hallucinations in Bengali LLM Responses

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Abstract

We describe our system for the *Olikbochon* Bengali LLM Hallucination Detection Challenge (IUT 12th ICT Fest 2026): given a Bengali prompt and a candidate answer, with or without a supporting passage, decide whether the answer is **faithful** or **hallucinated**. Rather than trusting a single language model, we build an *evidence-first, judge-second cascade*. A curated dataset of ≈105K verified question–answer references, exact hallucination pairs, an idiom dictionary, a symbolic math checker and BM25 retrieval resolve rows by exact evidence at near-100% precision wherever it applies; the remaining rows are decided by an ensemble of three complementary open-weight models — TigerLLM-9B, Qwen3.5-9B (also judged under an English cross-lingual frame) and a fine-tuned BanglaBERT encoder — fused by a per-track meta-judge fitted with cross-validation. Every evidence layer is precision-gated on the labeled set and self-disables if it errs. Our final system scores **0.886** public **/ 0.871** private macro-F1. Its design choices are independently corroborated by the organizers' BenHalluEval benchmark, which measured exactly the models and prompting strategies we adopt.

1 Introduction

Large language models fluently produce text that contradicts facts or the supporting context — *hallucination*. For Bengali, the sixth most spoken language, reliable hallucination detection is largely unstudied [1]. The Olikbochon challenge frames it as binary classification: each row is a triplet `(context, prompt, response)` where `context` is optional, and the goal is to predict $\mathrm{label} \in \{0 = \text{hallucinated}, 1 = \text{faithful}\}$. The scored metric is macro-F1, with the hallucinated class and a cultural-distance subset weighted in tie-breaking.

The decisive structural observation is that the negatives are *minimal-edit plausible distractors*: a single swapped name, year or number, or a literal↔figurative meaning swap. This makes *verifying the specific factual atom that changed* far more effective than any generic semantic-similarity score, and it shaped our whole design.

Contributions. We contribute (i) a **precision-gated evidence–judge cascade** whose exact-evidence layers settle every row they can before any model votes, each self-disabling if it errs; (ii) a **heterogeneous three-model ensemble** (Bengali-native LLM, multilingual LLM in Bengali and English frames, fine-tuned encoder) fused by a **per-track** meta-judge; (iii) a **benchmark-grounded routing policy** sending each task to its measured-best model; and (iv) an **efficiency redesign** (frozen-encoder inference, wall-time triage, batched generation, dual-GPU concurrency) that fits the 9-hour dual-T4 budget.

2 Task and Data

We are given 299 labeled and 2,516 unlabeled rows (Table 1). The decisive split is whether a `context` passage is present: *grounded* rows are a reading-comprehension *faithfulness* task (is the answer supported by the passage?), *closed-book* rows a world-knowledge *factuality* task with no evidence. Their optimal decision boundaries differ sharply — our encoder scores AUC 0.972 grounded vs. 0.571 closed — so every threshold and meta-judge is fitted *per track*.

Track	Definition	Labeled	Test
Grounded	context present	130	1,361
Closed-book	no context	169	1,155

Table 1: The grounded / closed-book split. Closed-book (world-knowledge) is the bottleneck and the target of most of our machinery.

3 Related Work

Prior hallucination benchmarks — TruthfulQA [6], FActScore [7], HaluEval [8] — share a labelling-bias blind spot: LLM judges default to one verdict regardless of content. **BenHalluEval** [1], the organizer-authored framework this competition follows, is the first dedicated Bengali hallucination benchmark: 12K GPT-generated hallucinated candidates seeded from TyDiQA-GoldP-Bengali [5], evaluating seven LLMs under a *dual-track* BenHalluScore that penalises both false positives (Track A) and missed hallucinations (Track B). We take its per-task model rankings as direct design evidence (Section 6) and use BanglaBERT [4] as encoder.

4 System Architecture

Our pipeline has three conceptual layers applied in ascending order of trust: an *evidence* layer (exact, rule-based), a *judge* layer (neural), and a *consistency* layer.

4.1 Evidence Layer (Exact-Match)

We assemble a **curated dataset** of ≈104,900 verified question→answer references, aggregated from public Bengali QA sources (reading-comprehension corpora, textbook QA, and public exam / general-knowledge sets), plus the benchmark's exact faithful/hallucinated answer pairs. Given a test row, the layer resolves it by *exact match* whenever possible, using echo-safe containment (a gold answer appearing in the response → faithful; a known distractor → hallucinated). Wherever exact evidence applies, the row is settled at 0.98–1.00 labeled precision and never reaches the judges.

Specialised rule-based checks handle recurring question types: a **symbolic checker** (SymPy) evaluates arithmetic prompts; a **field-aware idiom** module detects the literal↔figurative "swap trap" by matching the response against the *asked* meaning field; a **numeric-contradiction** rule flags a grounded response whose number is absent from the passage; and hand-authored **cultural-default (C1) traps** catch answers that match a predictable *global* default where the Bangladeshi answer differs (e.g. the national poet is Nazrul, not the reflexive Tagore; oral saline was developed in Bangladesh). Crucially, **every optional layer is precision-gated**: it is enabled only if it reaches ≥0.90 precision with sufficient support on the 299 labeled rows, and self-disables if its data is missing — making the system robust to the unseen Phase-2 fold.

4.2 Judge Layer (Neural)

Three heterogeneous open-weight judges, all 4-bit NF4-quantized to fit dual 16 GB T4 GPUs, provide complementary signals:

- **TigerLLM-9B** [2], a Bengali-native LLM, read via the first-token logit of the yes/no verbalizers — a single forward pass, no autoregressive decoding — giving a calibrated probability p_t.
- **Qwen3.5-9B** [3], a multilingual LLM (earlier: Qwen3-8B), scored twice — a Bengali frame (p_q) and an *English-instruction* frame (p_{qe}), since multilingual models judge more reliably in English while the content stays Bengali, recovering Bangladeshi facts they "know" in English space.
- **BanglaBERT** [4], fine-tuned *offline* on ≈13K real and ≈27K synthetic minimal-edit pairs, giving a fast encoder probability p_{clf}; loaded from a frozen checkpoint at inference and, being strong only on grounded rows, used there alone.

Each judge sees a two-shot prompt warning that "wrong answers are often a name/year/number slightly altered", targeting the minimal-edit distractors. Tiger (GPU 0) and Qwen (GPU 1) run in two threads, computing concurrently.

4.3 Per-Track Meta-Judge

For each track we fit and cross-validate three combiners — a two-way blend, a three-way simplex blend over $\{p_t, p_q, p_{qe}\}$, and a logistic-regression *stack* over all judge probabilities plus engineered features (p_{clf}, lexical Jaccard, response length, context coverage, the

Question type	Model combination (combiner; CV F1)
Grounded QA	Tiger + Qwen_{bn} + Qwen_{en} + BanglaBERT, logistic-regression stack (0.946)
Closed-book QA	$0.25\,$Tiger $+\,0.65\,\text{Qwen}_{\text{bn}}$ $+\,0.10\,\text{Qwen}_{\text{en}}$, 3-way blend (0.743)
Mathematical	SymPy verifier, else Qwen_{en} CoT vote — Tiger excluded

Table 2: Model combination per question type. The two neural QA tracks fuse three LLM judges — TigerLLM and Qwen in a Bengali (Qwen_{bn}) and an English (Qwen_{en}) frame — with the BanglaBERT encoder, with weights and thresholds set by per-track cross-validation. Math is routed to Qwen + SymPy with Tiger excluded (its measured weakness in Fig. 1). Rows resolved by exact evidence take no LLM.

numeric-contradiction flag) — and adopt the per-track winner, requiring a challenger to beat the blend by >0.005 CV macro-F1 to guard against overfitting 295 rows. The selection (Table 2) is intuitive: the *grounded* track picks the encoder-dominated stack (CV 0.946), the harder *closed* track a Qwen-heavy three-way blend; stack estimates use out-of-fold predictions to stay honest.

4.4 Decision Cascade and Consistency

At inference the meta-judge gives each row a provisional label; the evidence layers then override in ascending trust order (retrieval re-judge → math → reference containment → idiom / context / numeric tiers → curated answer keys → exact hallucination pairs → C1 traps → exact lookup). A final **consistency** layer makes identical (prompt, response) rows share a label and flags responses that contradict a *verified-faithful* anchor in their prompt group; a machine-readable audit logs every override.

5 Experiments and Results

Setup. All inference is local (dual T4, 4-bit models, disk $<$50 GB), within the 9-hour limit. Thresholds, weights and gates come from 5-fold CV on the labeled set; *we never tune on the leaderboard.*

Development journey. Table 3 traces ten successive approaches. From a multilingual zero-shot baseline (0.709), the single biggest model win was the Bengali-native judge (+0.04); recognising that the grounded contexts are public reading-comprehension paragraphs turned reference matching into an "answer key" (+0.03); a CV-gated two-judge blend and then a per-track meta-judge with a fine-tuned encoder lifted it further; and curated knowledge sets with dual-GPU judging reached our peak (0.897 public / 0.886 private). Two instructive *regressions* hardened standing rules: a learned decider that won CV but lost on the leaderboard taught us to gate on flip-*precision* not CV deltas; and a contaminated QA-verifier gate (its encoder was trained on data the validation rows came from) taught us to ask whether any gate's model was trained on the validation distribution.

Final result. The system scores **0.886** public **/ 0.871** private macro-F1. The labeled simulation (0.870 macro-F1, 0.855 hallucinated-class F1) matches the private leaderboard almost exactly — evidence the design transfers rather than overfits. Ablating the exact-evidence layers drops labeled macro-F1 to ≈0.78 (judge-only), confirming exact evidence is the largest single contributor.

6 Why the Design is Well-Founded

BenHalluEval's own measurements (Figure 1) corroborate our core choices, giving a direct comparison against named alternatives.

- **TigerLLM as primary judge:** BenHalluScore 28.28 on GQA, second of seven and ahead of every 27–32B multilingual model — the best *available* open judge for factual rows.
- **TigerLLM removed from math:** on reasoning it is the worst model (Track-B error 85.5%), so math is decided by Qwen and SymPy alone.
- **No CoT for factual judging:** the benchmark finds CoT helps only 9 of 21 model–task pairs (benefits confined to reasoning), so we judge by single-forward logits and reserve CoT for math.
- **Both error modes optimised:** since BenHalluScore punishes uniform-verdict

Approach	Pub.	Priv.	Key change / lesson
Baseline judge	.709	.732	multilingual zero-shot logit judge
Bengali judge	.753	.756	Bengali-native 3-track judge
Curated references	.789	.797	curated “answer-key” references + SymPy math verifier
Exact-pair refs	.797	.805	verified faithful / hallucinated answer pairs
Verifier grafts	.806	.797	BM25 retrieval + extractive-QA verification
Two-judge blend	.816	.815	CV-gated Tiger+Qwen ensemble + grammar references
Contamination fix	.818	.815	disabled a verifier whose gate leaked training data
Meta-judge	.843	.845	per-track stack + fine-tuned encoder + exact tiers
+ curated GK	**.897**	**.886**	curated knowledge sets + dual-GPU concurrent judging
Final (efficient)	**.886**	**.871**	frozen encoder checkpoint + wall-time triage

Table 3: Score ledger (public / private macro-F1) across successive approaches. The final submission is the efficient descendant of the highest-scoring line (Section 7).

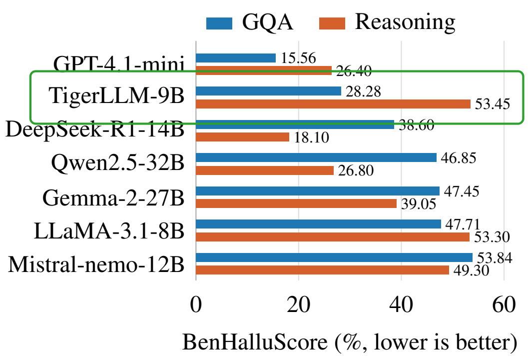

Figure 1: BenHalluScore (%, *lower is better*) across seven models from BenHalluEval [1] (green box: our judge). On **factual QA** — the task shape that dominates this competition — the 9B **TigerLLM** is second only to a closed API and beats every 14–32B multilingual model, so we make it our primary judge. On **reasoning** it is the *worst* model, so we route math to SymPy + Qwen instead.

bias, we tune macro-F1, refine the hallucinated-class threshold, and output a balanced 54/46 split with no verdict collapse.

Where the benchmark *evaluates* judges, we build the *detector* its findings imply — a per-track ensemble routed by these rankings.

7 Efficiency

Our top notebook fine-tuned the encoder *at runtime* ($\sim$18 min/rerun) and rebuilt references online; the submitted system keeps that accuracy while removing the cost. The encoder loads from a **frozen checkpoint**; **wall-time triage** lets tier-decided rows skip the judge passes; generation is **batched** ($\sim 7\times$); and the two judges run **concurrently** on both GPUs — cutting the judge block from $\approx$241 to $\approx$73 minutes. Reading verdicts as first-token logits saves an order of magnitude of tokens per row. We trade $\approx$1.5 private-LB points for a package that *provably completes and reproduces* on the held-out fold (50% of the grade).

8 Limitations

With only 299 labeled rows, several tiers auto-disabled for want of support — by design. The closed-book track is the ceiling ($\approx$0.74 CV): world-knowledge factuality without evidence is hard, and a portion of our Phase-1 coverage comes from lookup layers absent from the Phase-2 fold (per the organizers’ disclosure). The transferable core — judge ensemble, retrieval, grounded numeric verification — is register-agnostic and should carry that fold, but closed-book performance is the clearest headroom. All external data is public and cited; none derives from the test set.

9 Conclusion

We presented a precision-gated evidence–judge cascade for Bengali hallucination detection: exact evidence settles the rows it can at near-100% precision, and a benchmark-grounded per-track ensemble of three open-weight models decides the rest. It reaches 0.886 public / 0.871 private macro-F1, closely matching the labeled simulation — evidence it generalises rather than memorises.

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